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Quantum Kernels for EEG: An Invariance Mismatch Masquerading as Advantage

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Abstract

Objective. Quantum machine learning is often proposed for EEG decoding, but such claims rarely include the controls needed to show the quantum part is doing the work. We ask whether quantum kernels beat well-tuned classical baselines, and if not, why.

Approach. A trace-normalised EEG covariance matrix is a quantum density matrix, so quantum information geometry applies directly. We benchmark Hilbert–Schmidt, fidelity, Bures and quantum-relative-entropy kernels against tuned Riemannian, CSP and filter-bank baselines on two motor-imagery datasets under one nested cross-validation budget. An asymmetry then stands out: the quantum quantities are invariant only under orthogonal congruence, while EEG's nuisances span the full congruence group the classical baselines already handle. Borrowing recentring from Riemannian transfer learning, we refer each state to a training-set Fréchet mean, prove this restores exact affine invariance, and add a twin differing only in metric.

Main results. In the sensor frame the classical baselines won on both datasets. Recentring raised off-diagonal Gram variance several-fold and improved every kernel, by up to +0.106 on PhysioNet and +0.187 on IV-2a, the latter in every subject, reversing the headline comparison from +0.0523 in favour of the best classical pipeline to −0.0183 against it. The twin control shows this is not quantum. Across four settings spanning two datasets, two register sizes and both within- and cross-subject evaluation, two one-sided tests place the two families within ±0.032 accuracy of each other in all 20 comparisons, and within a pre-specified ±0.02 in 13 of them. Parity held against filter-bank CSP and in cross-subject transfer too.

Significance. We find no quantum advantage, and trace it to an invariance mismatch rather than kernel concentration: supply the missing invariance and the quantum kernels match their classical counterparts. Recentring is a precondition for a fair comparison, and omitting it is a concrete route to a false positive.

Keywords: quantum machine learning, brain–computer interface, electroencephalography, quantum kernels, Riemannian geometry, motor imagery, benchmarking

1 Introduction

Decoding motor imagery from EEG is a core problem in brain–computer interface (BCI) research, and the field has converged on a small set of strong classical methods: common spatial patterns (CSP) [1] and, more recently, classifiers built on the Riemannian geometry of spatial covariance matrices [2, 3]. Large-scale benchmarking finds that covariance tangent-space projection and CSP consistently define the strongest methodological families across public motor-imagery datasets; Lotte *et al* [4] and Yger *et al* [5] review the field and the Riemannian branch of it respectively.

In parallel, a growing literature applies quantum machine learning (QML) to EEG, typically reporting improvements over a classical comparator [6, 7]. These reports share a recurring methodological pattern that makes them difficult to interpret. Classical baselines are often weak or untuned: plain support vector machines on band power rather than the Riemannian or filter-bank CSP methods that define the state of the art. Hyperparameters are frequently optimised for the quantum model alone. Most critically, two controls are almost universally absent: an *entanglement ablation*, which tests whether the quantum resource contributes anything at all, and a *dimension-matched classical comparator*, without which an apparent quantum effect is confounded with the dimensionality reduction applied before encoding.

Quantum kernels are the natural object of study here. Supervised quantum models with fixed encodings are kernel methods [8, 9], so a kernel comparison captures the whole model class rather than one architecture, and it sidesteps the trainability problems that afflict variational circuits [10, 11]. What is known about their usefulness is mixed. There is a constructed problem with a proven exponential separation [12], but on natural data the advantage is bounded by how much the quantum kernel's inductive bias happens to match the target [13, 14], and general surveys [15] have not translated into demonstrated wins on biological signals.

The wider QML literature gives strong reasons for caution. Bowles *et al* [16] benchmarked twelve quantum models across 160 datasets and found that out-of-the-box classical models generally outperformed them; more pointedly, removing entanglement often left performance equal or better. Thanasilp *et al* [17] proved that quantum kernel values concentrate exponentially in qubit count toward a fixed value, collapsing the model to a trivial one. Neither result has been systematically tested on EEG.

This paper makes four contributions.

First, we identify a *principled* quantum encoding for EEG. An EEG trial's spatial covariance matrix is symmetric positive definite; normalised to unit trace it is a bona fide quantum density matrix (figure 1). This is an identity, not an analogy, and it makes quantum information geometry (Uhlmann fidelity, the Bures metric, von Neumann entropy) applicable to EEG natively, with no lossy squeezing of features into rotation angles. The Hilbert–Schmidt overlap $\mathrm{tr}(\rho\sigma)$ is moreover exactly the quantity a SWAP test estimates, so the construction maps onto a hardware-native primitive. Related tooling exists [18], but its quantum step operates on flattened tangent-space vectors, discarding the density-matrix structure.

Second, we report a controlled benchmark of 23 pipelines on 30 subjects with nested cross-validation, matched tuning budgets, both of the missing controls, and paired subject-wise statistics with multiple-comparison correction, replicated on IV-2a and extended to filter-bank baselines and cross-subject transfer.

Third, and centrally, we show that the comparison such benchmarks perform, ours included, in its first form, is confounded by the coordinate frame. The quantum quantities above are invariant under orthogonal congruence only, while EEG's nuisance transformations (electrode gain, reference choice, source mixing, and the change from one subject to the next) generate the full congruence group, under which every strong classical baseline *is* invariant. We prove that referring each state to a Fréchet-mean reference state restores exact affine invariance while remaining a legitimate quantum operation, and we show that this single change relieves concentration by 4.7–9.5× and reverses the sign of the classical-versus-quantum comparison on both datasets.

Fourth, we show that the reversal is nonetheless *not* evidence of quantum advantage, using a control we believe should be standard in this literature: a classical SPD-manifold kernel placed in the same classifier, on the same data, in the same frame, differing from the quantum kernels only in the metric. No quantum kernel is distinguishable from that twin in any of four settings. The practical implication is uncomfortable but concrete: a quantum kernel compared against a classical baseline in a different frame will show an advantage of roughly the size we report, produced entirely by the frame.

We state our scope plainly. At the channel counts considered here (8–64, i.e. 3–6 qubits) every kernel we evaluate is classically computable in $O(n_{\mathrm{ch}}^3)$. We make no claim of quantum speedup; the object of study is whether quantum *geometry* improves decoding. Every quantum kernel is evaluated by exact state-vector simulation: there is no gate error, no decoherence and no readout bias, and section 3.8 adds only sampling noise. Hardware would make the quantum side worse, never better, so the comparison throughout is maximally generous to it and a negative result here is not weakened by the idealisation.

2 Methods

2.1 Data and preprocessing

We use the PhysioNet EEG Motor Movement/Imagery Database (EEGMMIDB) [19, 20], recorded with the BCI2000 system at 160 Hz over 64 channels. We analyse the standard binary protocol: imagined left-hand versus right-hand movement (runs 4, 8 and 12), giving 45 trials per subject. Signals are band-pass filtered to 8–30 Hz, spanning the μ and β rhythms in which motor-imagery event-related desynchronisation appears, and epoched from 0.5 to 3.5 s after cue onset to exclude the cue-evoked transient. Epochs are resampled to 128 Hz.

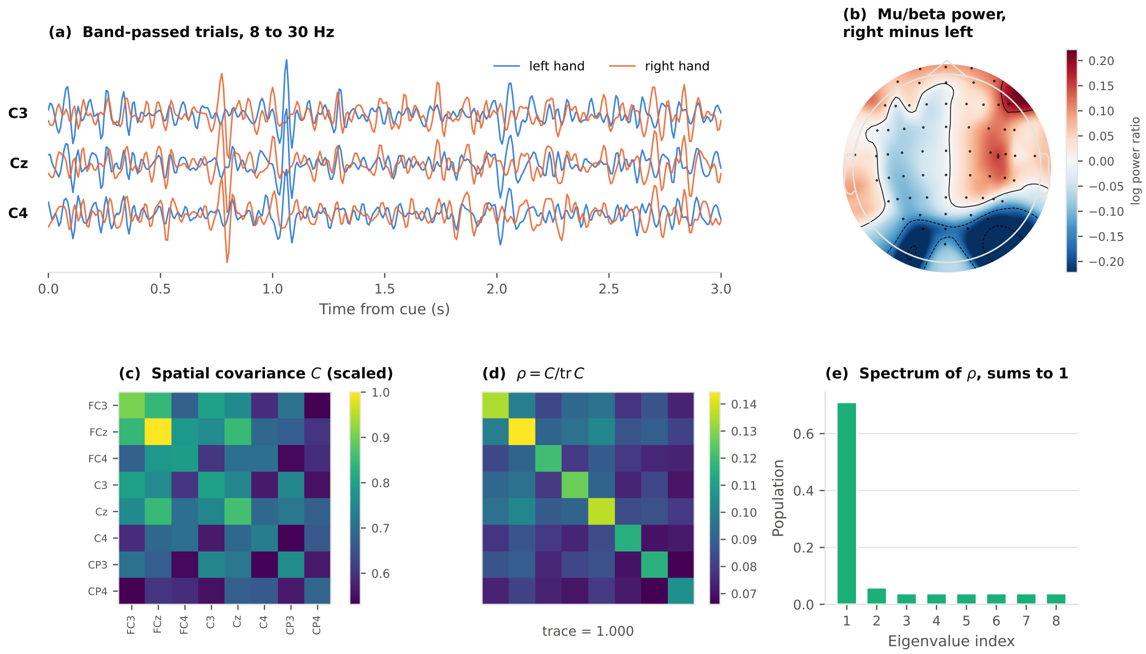

Figure 1. From scalp EEG to a quantum state, computed from real recordings rather than simulated. (a) Band-passed sensorimotor traces for one left-hand and one right-hand trial. (b) Scalp topography of mu/beta power, right minus left: power falls over the hemisphere contralateral to the imagined hand, which is the physiological effect every decoder in this study exploits. (c) The spatial covariance of a single trial over the eight decoding channels. (d) The same matrix divided by its trace. It is now a density matrix, and (e) its eigenvalues are non-negative and sum to one, so they are a population distribution. The step from (c) to (d) is the whole encoding: no dimensionality reduction, no rotation-angle embedding.

The primary analysis uses eight sensorimotor channels (FC3, FCz, FC4, C3, Cz, C4, CP3, CP4), a register of exactly three qubits under the amplitude encoding described below. Subjects 88, 89, 92, 100 and 104 are excluded for documented annotation and sampling-rate defects. We analyse the first 30 eligible subjects.

Spatial covariance matrices are estimated per trial with the oracle approximating shrinkage (OAS) estimator, which is well conditioned at the trial counts available here.

2.2 Second dataset: BCI Competition IV-2a

A single dataset cannot separate a property of the methods from a property of the data. PhysioNet gives breadth but only 45 trials per subject, so a negative result there invites the objection that the quantum models were starved. We therefore repeat the entire benchmark on BCI Competition IV-2a (9 subjects, 288 trials each, 2 025 fold scores), accessed through MOABB [21]. The two datasets sit at opposite corners of the design space: many subjects with little data each versus few subjects with a great deal.

IV-2a is also the dataset the quantum-EEG literature reports on [6], so results here are directly comparable to published claims. The same eight sensorimotor channels are used, all of which exist in the IV-2a montage, so the register size is identical at three qubits. Band-pass, resampling, pipelines, hyperparameter grids and cross-validation protocol are unchanged; only the data differs. Each dataset keeps its own standard epoch window, since each competition defines its own cue timing, and the comparison of interest is whether the *ranking* of pipelines moves rather than whether absolute accuracies coincide.

2.3 EEG covariance matrices as quantum states

Let $C_i \in \mathbb{R}^{n_{\mathrm{ch}} \times n_{\mathrm{ch}}}$ be the spatial covariance of trial i. Since C_i is symmetric positive definite,

$$\rho_i = \frac{C_i}{\operatorname{tr} C_i} \tag{1}$$

satisfies $\rho_i \succeq 0$ and $\operatorname{tr} \rho_i = 1$: it is a density matrix on an n_{ch}-dimensional Hilbert space, that is, on $\log_2 n_{\mathrm{ch}}$ qubits. Sweeping the channel count over powers of two therefore sweeps the register size with no change of method, a property we exploit in section 3.3.

2.4 Density-matrix kernels

Equation (1) makes the following quantum-information quantities directly available as kernels between trials.

The *Hilbert–Schmidt overlap*

$$K_{\mathrm{HS}}(\rho,\sigma)=\mathrm{tr}(\rho\sigma) \tag{2}$$

is an inner product on Hermitian matrices, so its Gram matrix is positive semi-definite by construction and requires no repair. It is precisely the observable estimated by a SWAP test.

The *Uhlmann fidelity* [22]

$$F(\rho,\sigma)=\left(\mathrm{tr}\sqrt{\sqrt{\rho}\,\sigma\sqrt{\rho}}\right)^2 \tag{3}$$

is the quantum generalisation of the Bhattacharyya coefficient. We evaluate it via the eigenvalues of the symmetrised product, which is numerically stable for the matrix sizes involved.

Both quantities proved severely concentrated on EEG (section 3.3), so we additionally define bandwidth-parameterised kernels that retain the quantum geometry while restoring a length scale, exactly as a radial basis function (RBF) kernel does for the Euclidean distance:

$$K_{\mathrm{HS\text{-}RBF}}(\rho,\sigma) = \exp\left(-\gamma\,\mathrm{tr}\left[(\rho-\sigma)^2\right]\right), \tag{4}$$

$$K_{\mathrm{Bures}}(\rho,\sigma) = \exp\left(-\gamma\, d_B^2(\rho,\sigma)\right), \qquad d_B^2=2\left(1-\sqrt{F(\rho,\sigma)}\right). \tag{5}$$

Here d_B is the Bures distance [23], the geodesic distance of the Bures metric on state space, which for unit-trace matrices coincides with the Bures–Wasserstein distance occasionally used in the Riemannian EEG literature as a distance but, to our knowledge, never as a kernel. The bandwidth γ is a bandwidth, and bandwidth is known to control quantum-kernel expressivity and to be decisive for whether such kernels generalise at all [24]. It is set by the median heuristic computed *on the training split only*, with a multiplier selected in the inner cross-validation loop.

2.5 Reference states, and the invariance the sensor frame lacks

Equation (1) expresses each state in the frame the electrodes happen to define. That frame is not neutral.

The nuisance transformations of EEG act on the spatial covariance by congruence, $C\mapsto ACA^{\mathsf{T}}$ with A invertible: per-electrode gain and impedance, the choice of reference electrode, linear source mixing and volume conduction, and the change from one subject or session to the next all take this form. The affine-invariant Riemannian metric is invariant under the whole congruence group, which is the structural reason tangent-space decoding is the classical state of the art and why it transfers.

The quantum quantities are not. $\mathrm{tr}(\rho\sigma)$, the Uhlmann fidelity, the Bures distance and the quantum relative entropy are invariant only under the *orthogonal* subgroup $A\in O(n_{\mathrm{ch}})$. This is an asymmetry in the comparison itself, not merely a property of the data: pyRiemann's tangent-space estimator fits a reference mean M and maps $C\mapsto\log(M^{-1/2}CM^{-1/2})$, and both the minimum-distance-to-mean classifier and CSP are congruence-invariant by construction. Every strong classical baseline in our suite therefore possesses an invariance that none of the sensor-frame quantum kernels possess.

The repair is not new, and we want to be plain about that. Whitening covariances by their Fréchet mean is *Riemannian recentring*, introduced by Zanini *et al* [25] to align sessions and subjects and extended by Rodrigues *et al* [26] into Riemannian Procrustes Analysis; He and Wu [27] give a Euclidean counterpart. In the classical setting it is understood as a domain-adaptation step, applied because it helps in transfer, and it is optional: the affine-invariant methods it is applied to are already congruence-invariant without it.

Our point is that for quantum kernels the same operation plays a different and stronger role. It is not an optional aid to transfer but the step that supplies an invariance the kernels structurally lack, and which the baselines they are being compared against already have. We therefore refer each state to a *reference state* M, the Fréchet mean of the *training* covariances, via the whitening $W=M^{-1/2}$:

$$\tilde{\rho}_i = \frac{WC_iW}{\mathrm{tr}\,(WC_iW)}. \tag{6}$$

Proposition. *Under a common congruence $C_i\mapsto AC_iA^{\mathsf{T}}$ with A invertible, the kernels $\mathrm{tr}(\tilde{\rho}\tilde{\sigma})$, $F(\tilde{\rho},\tilde{\sigma})$, $d_B(\tilde{\rho},\tilde{\sigma})$ and $J(\tilde{\rho},\tilde{\sigma})$ are invariant.*

Proof. The affine-invariant mean is equivariant, $M\mapsto AMA^{\mathsf{T}}$. Put $B=(AMA^{\mathsf{T}})^{-1/2}A$. Then $BMB^{\mathsf{T}}=I=WMW^{\mathsf{T}}$, so $U:=BM^{1/2}$ satisfies $UU^{\mathsf{T}}=I$ and $B=UW$. Hence

$WC_iW \mapsto U(WC_iW)U^{\mathsf{T}}$ for one common orthogonal U, and trace normalisation commutes with orthogonal conjugation. Each quantity above is invariant under a common unitary. □

In the reference frame the quantum kernels are thus *exactly* affine-invariant, sharing the invariance group of the affine-invariant Riemannian metric. Figure 2 sets out the geometry and verifies the invariance numerically. Two consequences follow, and we test both. First, the comparison against the classical baselines becomes like for like. Second, and less obviously, the quantum kernels can now at best *match* the Riemannian metric on any nuisance that is a congruence; parity, not superiority, is the ceiling the algebra permits.

To state the contribution precisely: the alignment is borrowed, the reason for needing it is not. What we add is the observation that quantum kernels lack an invariance the classical baselines already possess, a proof that recentring supplies exactly that invariance for the quantum quantities, and the control in section 3.5 showing that once it is supplied nothing quantum remains to distinguish them.

Equation (6) is not classical preprocessing smuggled into a quantum method. The map $\rho \mapsto W\rho W/\mathrm{tr}(W\rho W)$ is a filtering (Lüders) operation with Kraus operator W followed by renormalisation, so it is state preparation and the SWAP-test implementation path is unchanged. M is estimated on training data only.

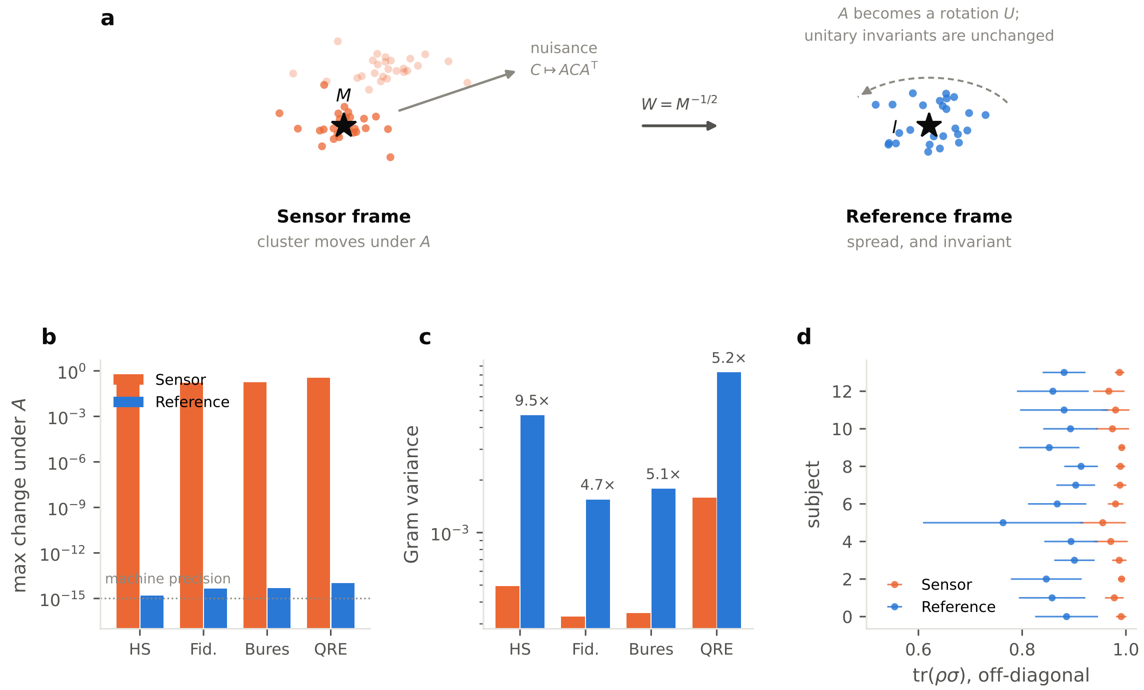

Figure 2. (a) EEG covariances cluster tightly about their Fréchet mean M, and a nuisance congruence moves the whole cluster. Whitening by M spreads the states about the identity and reduces the nuisance to a common rotation, to which every quantum invariant is blind. (b) The proposition verified numerically: reference-frame residuals sit at machine precision, sensor-frame ones are $O(10^{-1})$. (c) Off-diagonal Gram variance rises by 4.7–9.5× in the reference frame. (d) The sensor-frame overlaps saturate near unity in every subject ($n = 14$).

We also add the symmetrised quantum relative entropy

$$J(\rho, \sigma) = \frac{1}{2}\,\mathrm{tr}[(\rho - \sigma)(\log \rho - \log \sigma)]\,, \tag{7}$$

used as $\exp(-\gamma J)$. It is distinct from the classical Kullback–Leibler divergence between the Gaussians the covariances parameterise, which pyRiemann supplies as a distance and which involves $\mathrm{tr}(\sigma^{-1}\rho)$ and log-determinants rather than matrix logarithms.

2.6 Circuit embedding kernels and the entanglement ablation

For comparison with the prevailing approach, we also evaluate parameterised-circuit embedding kernels $K(x, z) = |\langle\phi(x)|\phi(z)\rangle|^2$ (figure 3), where ϕ is an instantaneous-quantum-polynomial (IQP) style feature map of the Havlíček *et al* family [28], built from Hadamard and R_Z rotations with data-dependent ZZ couplings. Inputs are Riemannian tangent-space features reduced by PCA to the number of qubits and scaled to a rotation-angle range; the angle scale is itself a tuned hyperparameter. A variant with fixed CNOT entanglers is also included.

The **entanglement ablation** is the decisive control: an otherwise identical circuit with every entangling gate deleted. The resulting kernel factorises into single-qubit kernels and is classically trivial. If it matches or exceeds the entangled version, quantum structure is not contributing [16].

Kernels are evaluated by preparing each embedded state once on a state-vector simulator, which is mathematically identical to running the adjoint kernel-estimation circuit at infinite shots but costs $O(n\,2^q)$ rather than $O(n^2)$ circuit executions.

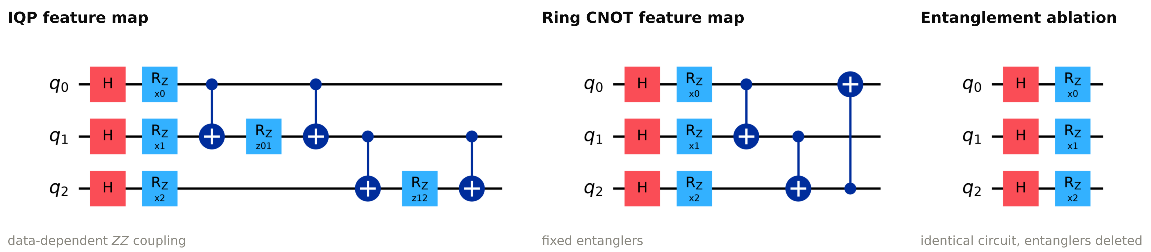

Figure 3. The circuit feature maps, at three qubits with one layer shown. Rotation angles carry the PCA-reduced tangent-space features and the kernel is the state overlap $|\langle\phi(x)|\phi(z)\rangle|^2$. The IQP map uses data-dependent ZZ couplings; the ring map uses fixed CNOTs. The third panel is the entanglement ablation: the identical circuit with every entangling gate deleted, which factorises the kernel into single-qubit terms and isolates what entanglement contributes. The figure is rendered by Qiskit from a circuit that is checked against the executed PennyLane circuit before drawing: the two agree on full Gram matrices to $< 10^{-15}$, so the diagram cannot drift from the code it documents.

2.7 Classical baselines and controls

Five classical baselines represent the genuine state of the art rather than a strawman: log-variance with shrinkage LDA; CSP with shrinkage LDA; minimum distance to Riemannian mean (MDM); Riemannian tangent space with logistic regression; and tangent space with an RBF SVM.

Four controls isolate specific confounds: the entanglement ablation described above; *dimension-matched* linear and RBF SVMs operating on exactly the same PCA features the circuit kernels receive; and a log-Euclidean tangent-space classifier as the classical geometric twin of the density-matrix kernels.

2.8 Evaluation protocol and statistics

Each subject is evaluated independently. The outer loop is a repeated stratified k-fold (5 folds $\times$ 3 repeats) estimating generalisation; the inner loop is a 4-fold grid search selecting hyperparameters. Crucially, *every* pipeline, including all classical baselines, receives the same inner search budget. All fitted transformations (tangent-space reference mean, scalers, PCA, and the kernel bandwidth) are refit within each training split, so no information leaks across folds.

Across subjects we follow the standard recommendation for comparing classifiers over multiple datasets [29] and use the Wilcoxon signed-rank test on paired per-subject accuracies, with Holm correction [30] across the family of comparisons, and report the paired effect size d_z. Per-subject optimality varies substantially in motor-imagery decoding, so paired non-parametric testing (rather than comparison of grand means) is the appropriate inference.

Two points about the correction structure, since several families appear below. Holm correction is applied *within* each family, and the families are the pre-specified ones: the suite-wide comparison against the reference pipeline, the pre-specified contrasts of table 3, and the metric-matched twin comparisons. These were fixed by the design of the study — one family per question asked — and not selected after inspecting results; no comparison reported here was chosen because it was significant. We deliberately do not apply a single global correction across every family, because the questions are logically distinct and a global correction would make the null results, which are the paper's actual claims, easier rather than harder to obtain.

Where the claim is that two methods are the *same*, a non-significant difference is not sufficient evidence, so we additionally report two one-sided tests (TOST) on the paired differences against a margin fixed in advance (table 9). The margin, its justification and the achieved equivalence bounds are given there.

2.9 Implementation

Analyses use MNE-Python [31], pyRiemann, scikit-learn and PennyLane [32]. Code and results are available at https://github.com/Imtiaj-Sajin/quantaEEG.

3 Results

Table 1. Within-subject decoding performance on PhysioNet EEGMMIDB left- versus right-hand motor imagery. Accuracy and AUC are means over subjects of the per-subject nested cross-validation score; SD is between subjects. Runtime is mean wall-clock seconds per subject for the full hyperparameter search. Rows are ordered by accuracy.

Pipeline	Group	Accuracy	SD	AUC	Runtime (s)
classical/CSP+LDA	Classical	**0.611**	0.154	0.636	1.99
classical/TS+LR	Classical	0.595	0.174	0.625	2.69
classical/TS+RBF-SVM	Classical	0.590	0.168	0.605	6.27
control/logeuclid-TS+LR	Control	0.582	0.149	0.611	2.65
control/PCA-matched-linear	Control	0.582	0.156	0.588	2.41
quantum/CNOT-kernel-SVM	Quantum	0.579	0.148	0.593	9.27
control/PCA-matched-RBF	Control	0.578	0.153	0.576	6.64
control/IQP-no-entangle	Control	0.576	0.147	0.594	8.26
classical/MDM	Classical	0.575	0.161	0.594	0.23
quantum/IQP-kernel-SVM	Quantum	0.566	0.134	0.588	10.19
classical/logvar+LDA	Classical	0.559	0.134	0.585	0.05
quantum/Bures-RBF-SVM	Quantum	0.552	0.109	0.567	12.40
quantum/HS-RBF-SVM	Quantum	0.549	0.101	0.567	5.85
quantum/HS-overlap-SVM	Quantum	0.532	0.076	0.578	2.09
quantum/Fidelity-SVM	Quantum	0.523	0.082	0.557	4.68

Table 2. Paired comparisons against the classical/TS+LR reference across subjects (Wilcoxon signed-rank, two-sided). Δ is the mean per-subject accuracy difference; d_z is the paired effect size; p_{Holm} is corrected across the whole family of comparisons. No comparison survives correction.

Pipeline	Δ accuracy	p	p_{Holm}	d_z	Better in
classical/CSP+LDA	+0.0168	0.156	1.000	+0.261	17/30
classical/TS+RBF-SVM	−0.0047	0.905	1.000	−0.105	16/30
control/logeuclid-TS+LR	−0.0121	0.405	1.000	−0.140	13/30
control/PCA-matched-linear	−0.0126	0.946	1.000	−0.165	18/30
quantum/CNOT-kernel-SVM	−0.0156	0.227	1.000	−0.217	11/30
control/PCA-matched-RBF	−0.0170	0.264	1.000	−0.242	9/30
control/IQP-no-entangle	−0.0183	0.316	1.000	−0.241	11/30
classical/MDM	−0.0195	0.198	1.000	−0.262	12/30
quantum/IQP-kernel-SVM	−0.0289	0.131	1.000	−0.316	13/30
classical/logvar+LDA	−0.0360	0.141	1.000	−0.283	11/30
quantum/Bures-RBF-SVM	−0.0425	0.081	0.978	−0.375	12/30
quantum/HS-RBF-SVM	−0.0452	0.111	1.000	−0.358	13/30
quantum/HS-overlap-SVM	−0.0625	0.048	0.628	−0.438	12/30
quantum/Fidelity-SVM	−0.0714	0.013	0.179	−0.498	10/30

Table 3. Pre-specified comparisons, first-named pipeline minus second. The entanglement ablation contrasts the IQP circuit kernel with an otherwise identical circuit whose entangling gates are deleted; the dimension-matched control contrasts it with a linear SVM on exactly the same PCA features. Only the primary classical-versus-quantum contrast is statistically significant; both ablations point in the expected direction but do not reach significance on accuracy at $n = 30$.

Comparison	Δ accuracy	p	d_z	Better in
Best classical vs. best quantum	+0.0323	**0.002**	+0.576	24/30
Entanglement ablation	+0.0106	0.309	+0.236	17/30
Dimension-matched control	+0.0163	0.173	+0.275	18/30

3.1 Quantum kernels do not outperform tuned classical baselines

Table 1 reports within-subject performance for all 15 pipelines; Figure 4 shows the same data with per-subject spread. Quantum kernels occupy five of the bottom six positions. Absolute accuracies are modest because EEGMMIDB contains many near-chance subjects, which is precisely why paired testing is required.

The primary pre-specified comparison, best classical versus best quantum, is significant: classical/CSP+LDA exceeded quantum/CNOT-kernel-SVM by $\Delta = +0.032$ ($p = 0.002$, $d_z = 0.58$), and was superior in 24/30 subjects (Table 3).

Table 4. Kernel concentration as a function of register size on real EEG. Register size is swept by varying the channel count over powers of two (4/8/16/32/64 channels = 2–6 qubits), with no change of method. The statistic is the variance of the off-diagonal Gram entries; a factor below one means concentration worsens with scale. The entangled circuit kernel concentrates 2.5× faster in log-slope than the same circuit with entanglers removed.

Kernel	Log-slope / qubit	Factor / qubit	Var (2 qubits)	Var (6 qubits)
IQP-entangled	−0.850	0.428	0.07824	0.00273
IQP-product	−0.334	0.716	0.09821	0.02633
Bures-RBF	−0.283	0.753	0.09509	0.03171
HS-RBF	−0.180	0.835	0.09318	0.04631
Fidelity	+0.369	1.447	0.00026	0.00125
HS-overlap	+0.419	1.520	0.00051	0.00356

Table 5. Replication on BCI Competition IV-2a (9 subjects, 288 trials each), with the PhysioNet result repeated for comparison. Protocol, channels, tuning budget and pipelines are identical; only the data differs. Quantum kernels occupy the four lowest positions on both datasets.

Pipeline	Acc (IV-2a)	AUC (IV-2a)	Acc (Phys.)
classical/TS+LR	**0.763**	0.822	0.595
classical/TS+RBF-SVM	0.755	0.811	0.590
classical/CSP+LDA	0.751	0.808	0.611
control/logeuclid-TS+LR	0.751	0.810	0.582
classical/MDM	0.723	0.777	0.575
classical/logvar+LDA	0.697	0.748	0.559
control/PCA-matched-linear	0.690	0.735	0.582
control/PCA-matched-RBF	0.687	0.729	0.578
quantum/CNOT-kernel-SVM	0.685	0.726	0.579
control/IQP-no-entangle	0.682	0.722	0.576
quantum/IQP-kernel-SVM	0.672	0.712	0.566
quantum/Bures-RBF-SVM	0.600	0.630	0.552
quantum/HS-RBF-SVM	0.595	0.621	0.549
quantum/HS-overlap-SVM	0.586	0.629	0.532
quantum/Fidelity-SVM	0.580	0.616	0.523

Table 6. Pre-specified comparisons on both datasets, combined by Fisher's method. Positive Δ favours the first-named pipeline. At $n = 9$ the two-sided signed-rank test cannot return $p < 0.0039$, so the IV-2a entries marked † sit at that floor: the test is saturated rather than merely significant. Both classical-versus-quantum contrasts hold in every one of the nine IV-2a subjects.

Comparison	Dataset	Δ acc	p	Better
Best classical vs. best quantum	PhysioNet	+0.0323	0.002	24/30
	IV-2a	+0.0662	0.004†	9/9
	Fisher combined		**<0.001**	
Entanglement ablation	PhysioNet	+0.0106	0.309	17/30
	IV-2a	+0.0100	0.203	7/9
	Fisher combined		0.237	
Dimension-matched control	PhysioNet	+0.0163	0.173	18/30
	IV-2a	+0.0178	0.129	7/9
	Fisher combined		0.107	

Table 7. Effect of referring the density-matrix kernels to a reference state. Each kernel is evaluated twice under an identical protocol, differing only in whether the states are expressed in the sensor frame ((1)) or relative to the training-set Fréchet mean ((6)). Δ is the mean per-subject accuracy gain from the reference frame, tested by paired Wilcoxon signed-rank across subjects. Every kernel improves on both datasets; on IV-2a every kernel improves in every subject.

Dataset	Kernel	Sensor	Reference	Δ	p	Better
PhysioNet ($n = 30$)	Fidelity	0.523	0.630	**+0.1064**	<0.001	23/30
	HS overlap	0.532	0.621	**+0.0889**	0.002	21/30
	HS-RBF	0.549	0.606	**+0.0570**	0.049	19/30
	Bures-RBF	0.552	0.611	**+0.0585**	0.009	20/30
	QRE-RBF	0.559	0.609	**+0.0504**	0.027	20/30
IV-2a ($n = 9$)	Fidelity	0.580	0.767	**+0.1873**	0.004	**9/9**
	HS overlap	0.586	0.762	**+0.1760**	0.004	**9/9**
	HS-RBF	0.595	0.763	**+0.1678**	0.004	**9/9**
	Bures-RBF	0.600	0.765	**+0.1646**	0.004	**9/9**
	QRE-RBF	0.599	0.764	**+0.1644**	0.004	**9/9**

Table 8. The decisive control. Each quantum kernel is compared against an SPD-manifold kernel used in the same support vector machine, on the same covariances, with the same tuning budget, in the same reference frame: only the metric differs. The comparator is the stronger of the two classical kernels in each setting, which is the conservative choice. Columns give the best quantum kernel in that setting, the classical twin, their difference, and the smallest p obtained by *any* of the five quantum kernels against the twin. No quantum kernel is distinguishable from its classical twin in any setting.

Setting	n	Quantum	Twin	Δ (range over 5)	$\min p$
PhysioNet, 3 q	30	0.630	0.621	$[-0.0151, +0.0081]$	0.183
PhysioNet, 5 q, filter bank	30	0.648	0.637	$[+0.0000, +0.0111]$	0.234
PhysioNet, transfer (LOSO)	30	0.647	0.635	$[+0.0000, +0.0119]$	0.072
BCI IV-2a, 3 q	9	0.767	0.766	$[-0.0033, +0.0016]$	0.426

Table 9. Equivalence of each quantum kernel to its classical twin, by two one-sided tests on the paired per-subject differences. A non-significant difference would only be an absence of evidence; TOST instead takes a *difference* as its null, so rejecting it supports equivalence. The margin $m = 0.02$ accuracy was fixed in advance and is smaller than the weakest effect this paper claims as real (the reference-frame correction is worth +0.050 to +0.187). CI is the 90 % interval, which corresponds to TOST at $\alpha = 0.05$; "bound" is the smallest margin at which equivalence would hold, so a reader preferring a stricter margin can read the answer off directly.

Setting	Kernel	n	Δ	90 % CI	p_{TOST}	Equiv.	Bound
PhysioNet, 3 qubits	Fidelity	30	+0.0081	$[-0.0042, +0.0205]$	0.0568	—	0.0205
	HS overlap	30	−0.0005	$[-0.0130, +0.0120]$	0.0066	✓	0.0130
	HS-RBF	30	−0.0151	$[-0.0317, +0.0016]$	0.3090	—	0.0317
	Bures-RBF	30	−0.0109	$[-0.0255, +0.0038]$	0.1487	—	0.0255
	QRE-RBF	30	−0.0121	$[-0.0267, +0.0025]$	0.1833	—	0.0267
PhysioNet, 5 qubits	Fidelity	30	+0.0111	$[-0.0045, +0.0268]$	0.1711	—	0.0268
	HS overlap	30	+0.0022	$[-0.0148, +0.0192]$	0.0431	✓	0.0192
	HS-RBF	30	+0.0000	$[-0.0159, +0.0159]$	0.0204	✓	0.0159
	Bures-RBF	30	+0.0017	$[-0.0111, +0.0145]$	0.0109	✓	0.0145
	QRE-RBF	30	+0.0015	$[-0.0125, +0.0155]$	0.0163	✓	0.0155
Transfer (LOSO)	Fidelity	30	+0.0007	$[-0.0097, +0.0112]$	0.0020	✓	0.0112
	HS overlap	30	+0.0119	$[+0.0002, +0.0235]$	0.1231	—	0.0235
	HS-RBF	30	+0.0000	$[-0.0154, +0.0154]$	0.0175	✓	0.0154
	Bures-RBF	30	+0.0052	$[-0.0080, +0.0183]$	0.0327	✓	0.0183
	QRE-RBF	30	+0.0067	$[-0.0068, +0.0201]$	0.0514	—	0.0201
BCI IV-2a, 3 qubits	Fidelity	9	+0.0016	$[-0.0049, +0.0080]$	0.0004	✓	0.0080
	HS overlap	9	−0.0033	$[-0.0109, +0.0043]$	0.0018	✓	0.0109
	HS-RBF	9	−0.0028	$[-0.0084, +0.0028]$	0.0002	✓	0.0084
	Bures-RBF	9	−0.0008	$[-0.0073, +0.0058]$	0.0003	✓	0.0073
	QRE-RBF	9	−0.0019	$[-0.0081, +0.0043]$	0.0003	✓	0.0081

Table 10. Cross-subject transfer, leave-one-subject-out: each model is trained on the pooled trials of every other subject. Hyperparameters are selected by subject-grouped inner cross-validation on the training subjects only. In the reference frame each subject is whitened by its own Fréchet mean, which uses no labels and is therefore legitimate unsupervised adaptation for the held-out subject. Δ is the gain from the reference frame, paired by held-out subject. Every method improves; in the reference frame no method is distinguishable from any other.

Pipeline	Sensor	Reference	Δ	p	Better
quantum/HS-overlap	0.552	0.647	+0.0948	<0.001	24/30
classical/TS+LR	0.604	0.644	+0.0400	0.010	18/30
quantum/QRE-RBF	0.576	0.641	+0.0659	0.002	21/30
quantum/Bures-RBF	0.561	0.640	+0.0785	<0.001	22/30
quantum/Fidelity	0.559	0.636	+0.0763	<0.001	21/30
control/riemann-kernel-SVM	0.587	0.635	+0.0474	0.004	19/30
quantum/HS-RBF	0.564	0.635	+0.0711	0.002	20/30
control/logeuclid-kernel-SVM	0.576	0.633	+0.0570	0.001	20/30
classical/MDM	0.553	0.633	+0.0800	<0.001	19/30

Table 11. Accuracy under finite-shot estimation. $\mathrm{tr}(\rho\sigma)$ is the SWAP-test observable, so S shots give an unbiased estimate with variance $(1-k^2)/S$; we sample each unordered pair binomially and project the Gram matrix back to the positive semi-definite cone. The reference frame needs *more* shots to approach its own ceiling, because that ceiling is higher, but it dominates the sensor frame in absolute terms from 10^4 shots upwards, above which it exceeds what the sensor frame achieves with unlimited shots.

Kernel	Frame	10^2	10^3	10^4	10^5	10^6	∞
HS-RBF	Sensor	0.492	0.504	0.511	0.538	0.533	**0.554**
HS-RBF	Reference	0.482	0.529	0.564	0.586	0.590	**0.609**
HS-overlap	Sensor	0.510	0.507	0.532	0.538	0.527	**0.532**
HS-overlap	Reference	0.491	0.518	0.565	0.589	0.607	**0.618**

Against the Riemannian tangent-space reference (Table 2, Figure 5), 0 of 14 comparisons survived Holm correction. Only two reached $p < 0.05$ uncorrected, and in both the quantum model was *worse*. No quantum kernel exceeded the reference at any significance level.

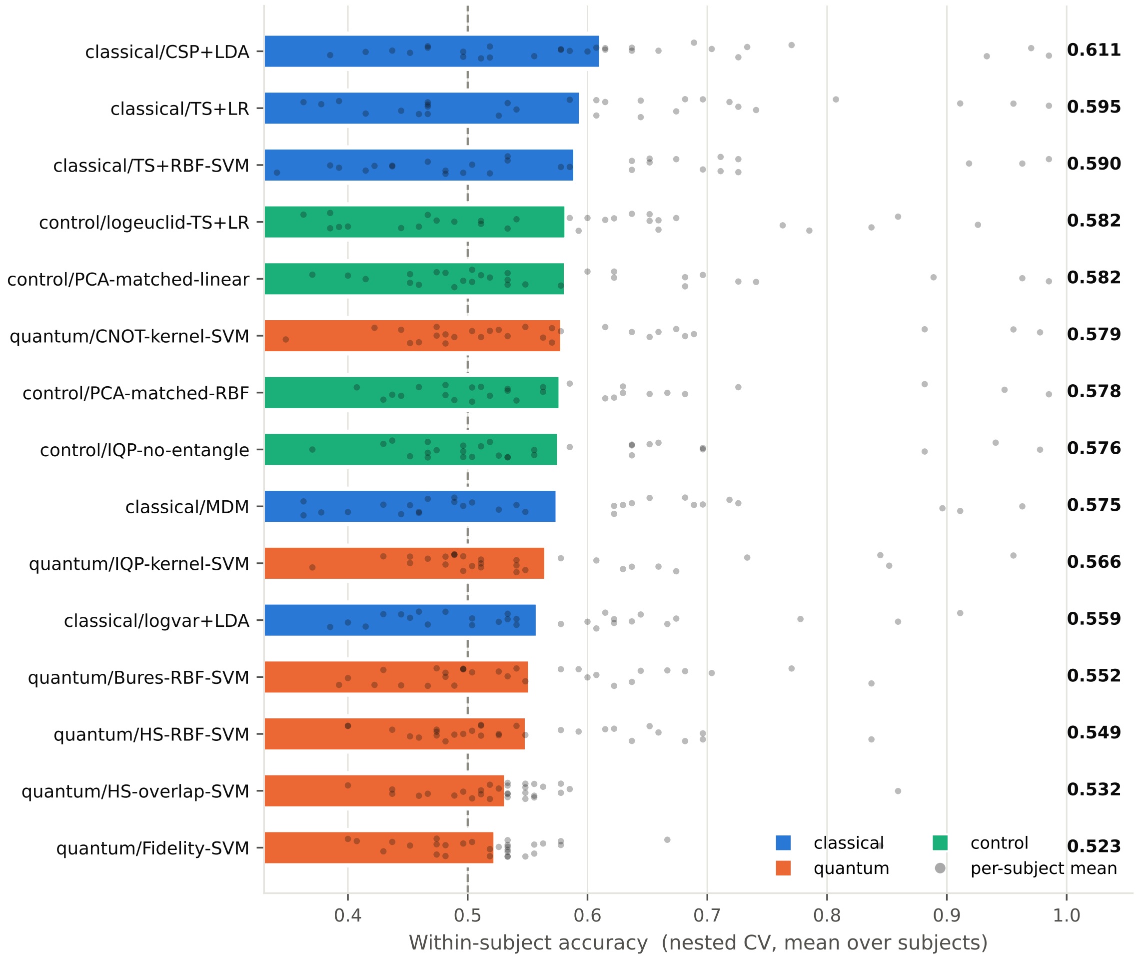

Figure 4. Within-subject accuracy by pipeline ($n = 30$ subjects, nested cross-validation). Bars are means over subjects; grey dots are individual subject means. The dashed line marks chance. Every pipeline, classical baselines included, received the same inner hyperparameter search.

3.2 The controls: entanglement and dimension matching

Both control ablations point in the direction predicted by [16], but neither reaches significance on accuracy (Table 3). Removing entanglement changed accuracy by +0.0106 ($p = 0.309$, superior in 17/30 subjects); the dimension-matched linear SVM exceeded the IQP circuit kernel by +0.0163 ($p = 0.173$).

We regard the honest reading as follows: deleting the quantum resource did not harm performance and numerically helped, but accuracy alone, measured on a noisy and low-SNR signal, does not have the power to establish the effect. The next section shows that the same effect is unambiguous when measured on the kernel itself.

3.3 Kernel concentration, and why entanglement makes it worse

Raw overlap kernels are severely concentrated on EEG. Off-diagonal Gram entries had mean 0.991 ± 0.010 (Hilbert–Schmidt) and 0.988 ± 0.010 (fidelity): every pair of trials appears $\approx 99\%$ identical, the Gram matrix is effectively rank-one, and the classifier has nothing to separate. This is the pathology proved by Thanasilp *et al* [17], here reproduced on real biological data rather than synthetic constructions.

Exploiting (1), we swept register size by varying the channel count over powers of two (4/8/16/32/64 channels = 2–6 qubits) with no change of method (Table 4, Figure 6). Two findings emerge.

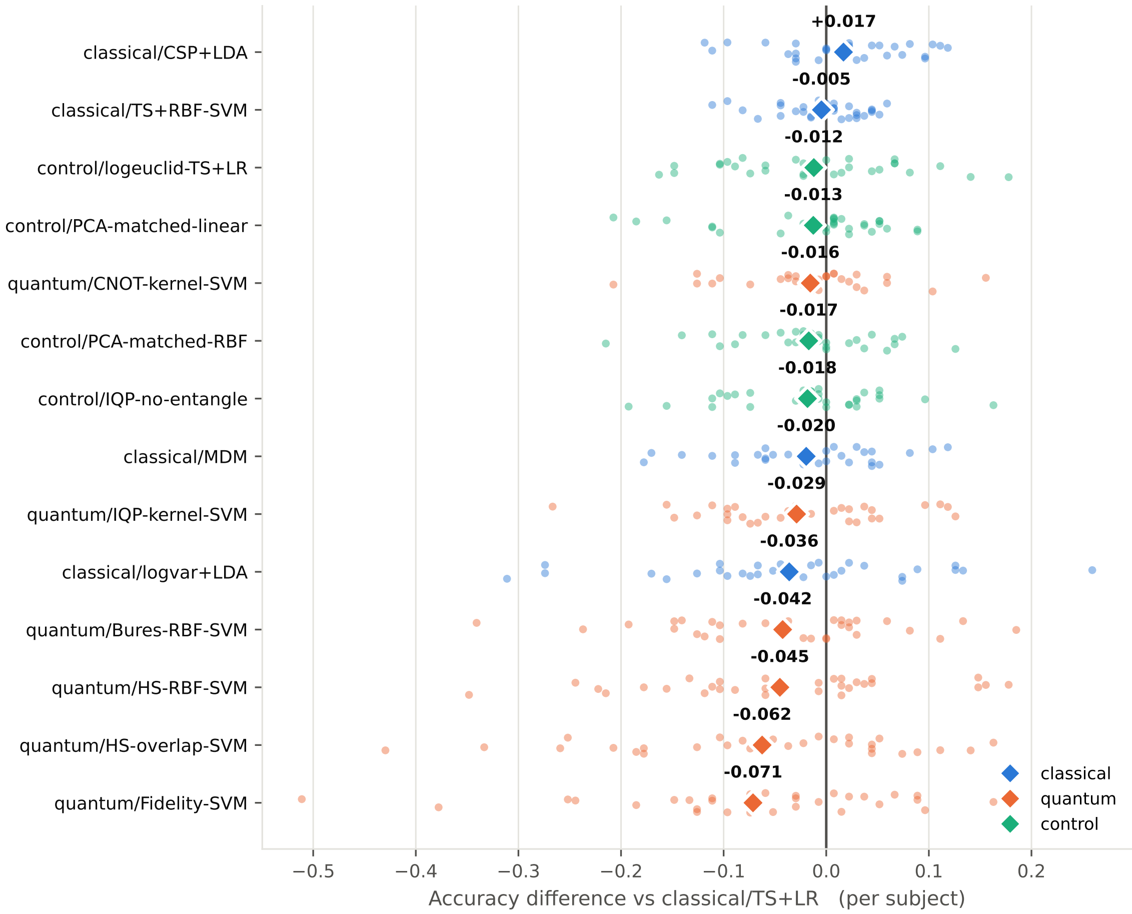

Figure 5. Paired per-subject accuracy differences against the Riemannian tangent-space reference. Dots are individual subjects; diamonds are means. Points left of zero indicate worse performance than the reference.

First, **entanglement accelerates the collapse**. The entangled IQP kernel loses variance by a factor of 0.428 per qubit, more than halving with each added qubit, while the identical circuit with entanglers deleted decays at 0.716. In log-slope the entangled kernel concentrates 2.5× faster (-0.850 versus -0.334 per qubit), a 29× versus 3.7× loss across the range. This supplies a mechanism for the observation of [16] that removing entanglement often improves QML models: entanglement drives embedded states toward mutual orthogonality, pushing the Gram matrix toward the identity, a memorising, non-generalising model.

Second, **the two kernel families concentrate for different reasons**. The raw density-matrix kernels do not follow the qubit-count law at all; their variance *increases* with dimension (factors 1.45–1.52). They are already saturated at two qubits because EEG covariance matrices are intrinsically similar to one another, a property of the data, not of the register. The practical corollary inverts standard practice: the density-matrix route should use *more* channels, not fewer.

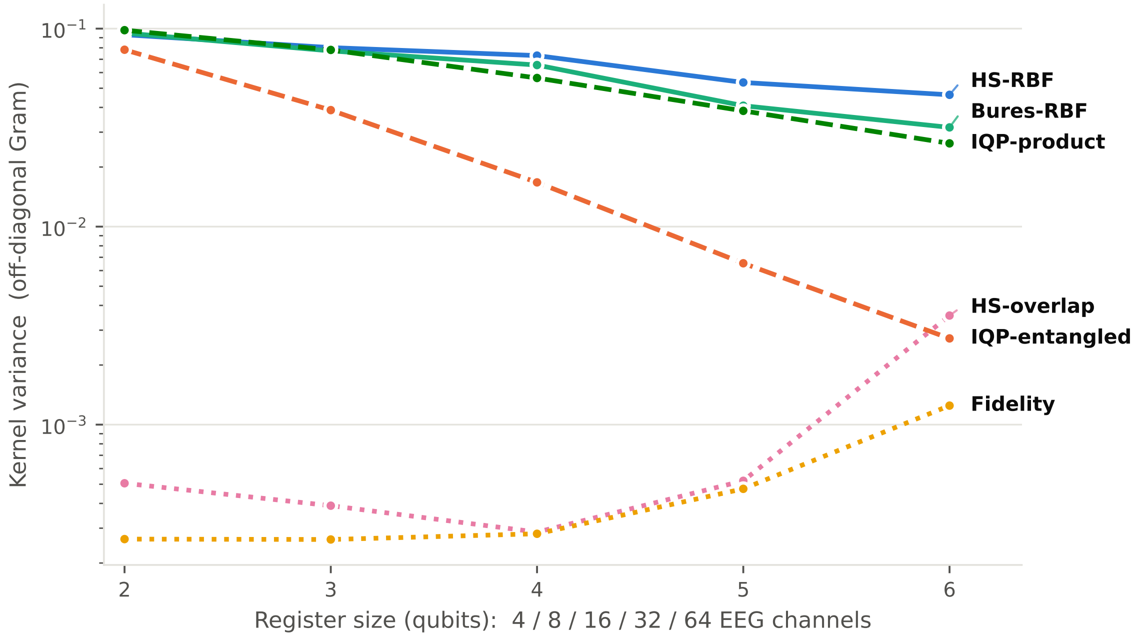

Figure 6. Kernel variance versus register size on real EEG ($n = 10$ subjects). Lower is worse: a kernel whose variance collapses cannot separate trials. Line style encodes the kernel family. The entangled circuit kernel (orange, dashed) decays fastest.

3.4 Replication on IV-2a: more data widens the gap

Repeating the entire benchmark on BCI Competition IV-2a reproduces the ordering and sharpens it (Table 5, Figure 7). The best classical pipeline reached 0.763 against 0.685 for the best quantum kernel and 0.580 for the worst, a classical-versus-quantum gap of 0.079 where PhysioNet gave 0.032, and a gap to the weakest quantum kernel of 0.183 against 0.088.

The direction of that change is the point. Classical methods gained roughly 0.15 accuracy from the six-fold increase in trials; the density-matrix kernels gained about 0.05. **More and cleaner data made the quantum kernels relatively worse, not better**, which is the opposite of what a data-starvation explanation predicts, and exactly what section 3.3 implies: a kernel whose off-diagonal entries all sit near unity gains no additional structure from additional trials, while the classical methods gain a great deal.

The ranking is also stable. Spearman rank correlation between the two orderings is 0.882 across all 15 pipelines, and the four lowest positions are occupied by the same four quantum kernels on both datasets with zero rank shift. This is a property of the methods, not of the dataset.

Statistically (Table 6), on IV-2a the best classical pipeline beat the best quantum kernel by $+0.066$ ($p = 0.004$, $d_z = 1.14$) in 9/9 subjects, and beat the density-matrix kernel by $+0.171$ ($d_z = 1.45$) in 9/9 subjects. Combining the two independent datasets by Fisher's method gives $p < 0.001$ and $p < 0.001$ respectively.

One caveat must be stated precisely rather than left to the reader. With n pairs the two-sided signed-rank test cannot return a p below 2^{1-n}; at $n = 9$ that floor is 0.0039, so a Holm correction over 14 comparisons cannot fall below 0.0547 whatever the effect size. Ten of the fourteen IV-2a family comparisons sit exactly at that floor. Reporting them as "not significant after correction" would misdescribe a saturated test as a null result, which is why the pre-specified comparisons in Table 6, which require no family correction, carry the inference.

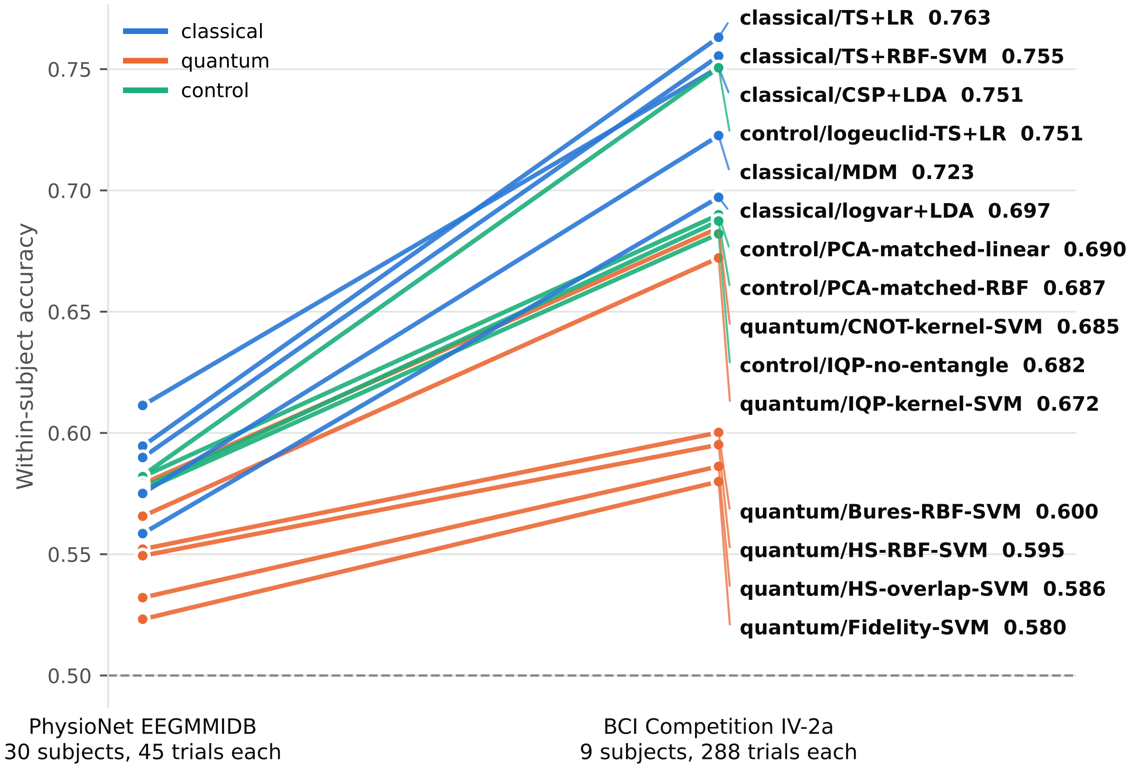

Figure 7. The same 15 pipelines on two datasets at opposite corners of the design space. Identical channels, protocol, tuning budget and code; only the data differs. Line slope is the quantity of interest: classical and control pipelines rise steeply with the extra trials while the four density-matrix and overlap kernels stay low. The dashed line marks chance.

3.5 The sensor frame is a confound, and correcting it reverses the result

Every result above evaluates the density-matrix kernels in the sensor frame. Section 2.5 argued that this is not neutral: the quantum kernels are invariant only under orthogonal congruence, while the tangent-space, MDM and CSP baselines they are being compared against are invariant under the full congruence group that EEG's nuisances generate. We now re-run the entire benchmark with the density-matrix kernels expressed relative to a training-set reference state ((6)), changing nothing else. The core pipelines reproduce their published values to ≤ 0.0001, so the two suites are directly comparable.

Three consequences follow, and the third is the one that matters.

First, concentration is largely an artefact of the frame. Measured over 14 subjects, the off-diagonal variance of the Gram matrix rises by a factor of 4.7–9.5 on moving to the reference frame. The sensor-frame Hilbert–Schmidt kernel has off-diagonal mean 0.981 and standard deviation 0.022; a nuisance congruence of the kind an electrode-gain change produces shifts individual kernel entries by an order of magnitude more than that entire discriminative spread. The kernel was measuring the frame, not the trial.

Second, accuracy improves substantially and consistently (table 7). On PhysioNet every kernel improves, by +0.050 to +0.106 ($n = 30$, all $p \leq 0.049$). On IV-2a the same corrections are worth +0.164 to +0.187, and every kernel improves in *every* subject. That the effect is two to three times larger on IV-2a is expected rather than surprising: the reference state is a Fréchet mean estimated from training covariances, and 288 trials per subject estimate it far better than 45 do.

This also revises the interpretation of section 3.4. There, more data widened the classical–quantum gap, which we read as evidence that the quantum kernels could not exploit additional trials. The narrower and correct reading is that a kernel with the wrong invariance cannot exploit them. Given the right frame, the extra data is not merely usable but more valuable to the quantum kernels than to the classical baselines.

The headline comparison consequently reverses. Against classical/CSP+LDA, the best sensor-frame quantum kernel loses by +0.0523 ($p = 0.016$, classical better in 20/30 subjects); the best reference-frame kernel, quantum/Fidelity-ref-SVM, instead leads by $|-0.0183|$ ($p = 0.059$, classical better in only 10/30). This does not reach conventional significance, and we do not claim it as an effect. We dwell on it because it is the size and shape of result that this literature routinely reports as a positive finding, and because everything needed to present it as one — two datasets, a consistent direction, a larger margin on the better-powered data — is in hand at this point in the analysis.

Third, it is not a quantum advantage, and the control that establishes this is the central result of this paper. A reference-frame quantum kernel differs from a classical Riemannian kernel in exactly one respect: the metric. Both consume the same covariances, both are used as precomputed kernels in the same support vector machine, both receive the same tuning budget, and after (6) both are affine-invariant. Table 8 reports that comparison in four independent settings: two datasets, two register sizes, and within- versus cross-subject evaluation.

In none of them is any quantum kernel distinguishable from its classical twin (figure 9). On PhysioNet the differences span -0.0151 to +0.0081 with a smallest p of $= 0.183$ in that setting; the twin itself, control/riemann-kernel-SVM, reaches 0.621 and is the second-best pipeline of the whole extended suite. The apparent quantum advantage is therefore attributable to two things, neither of them quantum: expressing the states in the reference frame, and using an SPD-manifold kernel in a support vector machine rather than a tangent-space projection.

A non-significant difference is only an absence of evidence, and the claim we are making deserves better than that, so we test equivalence directly (table 9). Two one-sided tests on the paired differences, with a margin of ± 0.02 accuracy fixed in advance, establish equivalence in 13 of 20 comparisons; across all 20 the agreement is within ± 0.032. The evidence is strongest exactly where it should be, on the better-powered dataset: on IV-2a all 5 comparisons are equivalent, with the widest interval reaching only 0.0109.

Two honest qualifications. First, the 4 comparisons that miss the ± 0.02 margin in the quantum kernel's favour are balanced by 3 that miss it in the twin's, which is the pattern expected of noise about zero rather than of a suppressed effect. Second, exactly 1 of the 20 intervals excludes zero at all: HS overlap (Transfer (LOSO)), at +0.0119. We do not read that as an advantage — it is one comparison of 20, it does not survive the family, and its interval still lies inside ± 0.032 — but it is the single place in this study where a quantum kernel is nominally ahead of a metric-matched classical control, and a reader is entitled to see it named.

We stress the counterfactual. Without this control, the same experiments support the claim that quantum kernels outperform tuned classical baselines at $p = 0.059$, on two datasets, with the gap widening on the better one. That claim would have been false. It is also, we suggest, the mechanism by which favourable results are obtained elsewhere in this literature: a quantum kernel that operates on whitened or aligned covariances, compared against a classical baseline that does not, will show an advantage of roughly the size reported in table 7 that has nothing to do with quantum structure.

3.6 Filter-bank baselines and a larger register

Since the reference-frame kernels reach the top of the table, the relevant question becomes whether they do so against the strongest classical family rather than against CSP and tangent-space alone. We therefore repeated the comparison with a filter bank of four sub-bands spanning the same passband, in the manner of filter-bank CSP [34], giving filter-bank CSP and filter-bank tangent-space baselines. The same construction enlarges the quantum register at no methodological cost: four sub-bands over eight channels is a 32-dimensional covariance, hence a single density matrix on five qubits rather than three.

Parity survives. The best quantum kernel reaches 0.648 against 0.627 for filter-bank CSP, a difference of +0.0212 ($p = 0.125$), and the classical-twin control replicates at this register size as well (table 8). The frame effect is larger here than at three qubits, +0.124 to +0.152; in the sensor frame the five-qubit kernels fall to chance and are the only comparisons in that family to survive Holm correction, in the unfavourable direction.

3.7 Cross-subject transfer

Within-subject accuracy is not the field's bottleneck; transfer to a new user is. It is also the setting in which a different geometry is most often conjectured to help. Table 10 and figure 10(a) report leave-one-subject-out evaluation over 30 subjects, training on the pooled trials of all others.

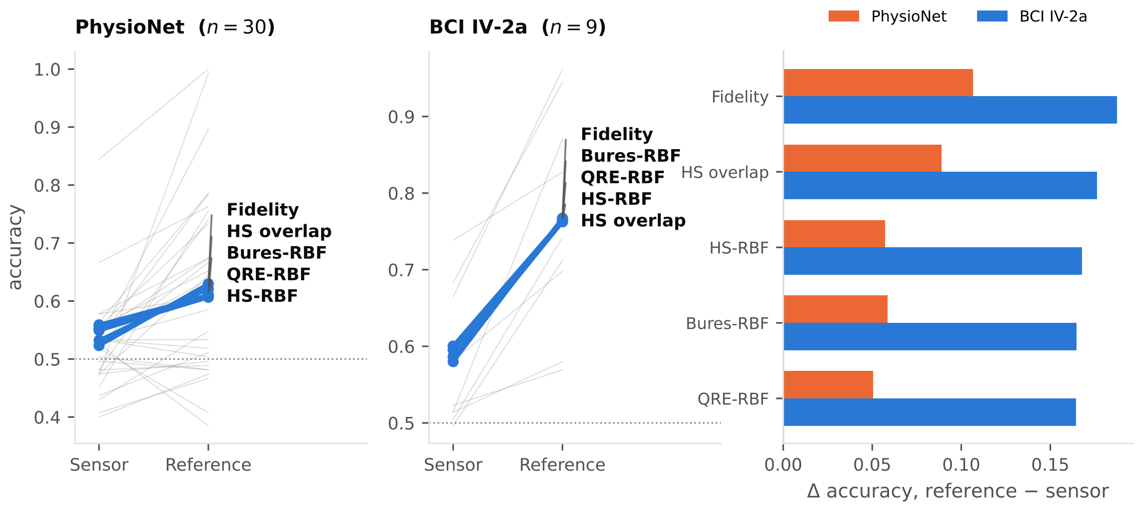

Figure 8. Accuracy in the sensor frame versus the reference frame. Thin lines are individual subjects, shown for the Fidelity kernel only. Every kernel improves on both datasets; on IV-2a every kernel improves in every subject. The correction is worth two to three times more on IV-2a, where the reference state is estimated from 288 rather than 45 trials per subject.

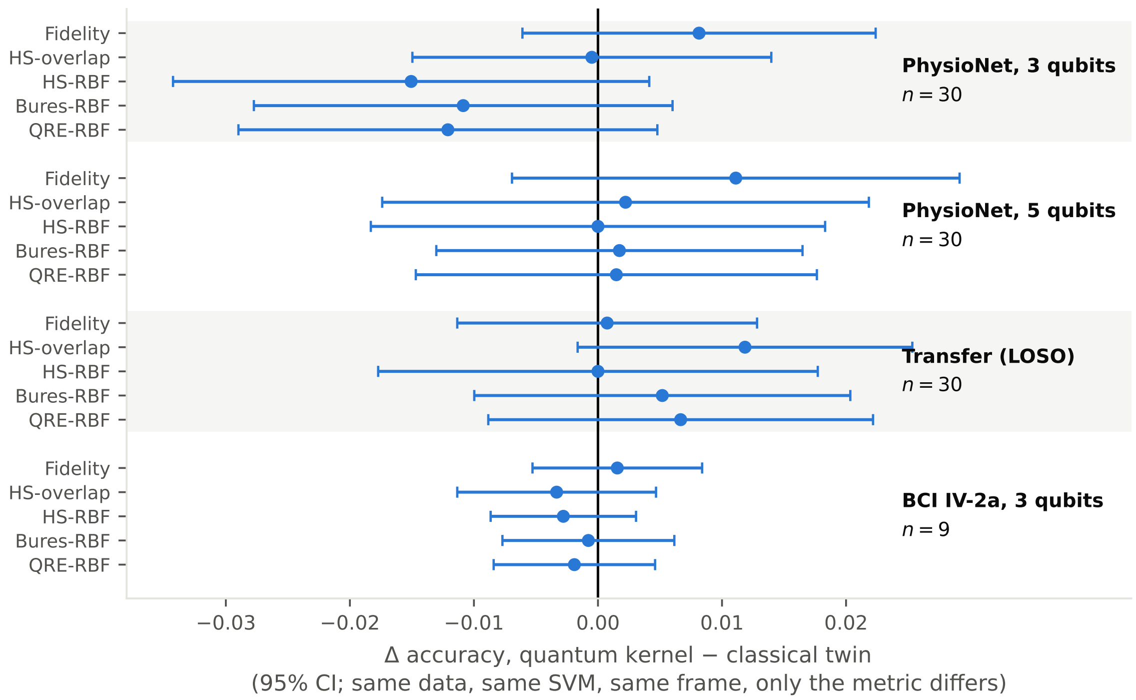

Figure 9. The decisive control. Each quantum kernel is compared against a classical SPD-manifold kernel [33] used in the same support vector machine, on the same covariances, with the same tuning budget, in the same reference frame; only the metric differs. Points are mean paired differences with 95% confidence intervals. Every interval crosses zero in all four settings. Note the axis range: the entire plot spans eight accuracy points, against a frame effect (figure 8) of up to nineteen.

The invariance argument makes a sharp prediction here. Inter-subject shift is dominated by congruence, so in the sensor frame the quantum distances must be worse than the affine-invariant Riemannian one, and in the reference frame they can at best equal it: parity is the ceiling the algebra permits, and no experiment can exceed it. Both halves hold. Every method gains from recentring, the non-invariant quantum kernels gaining most; and in the reference frame the best method, quantum/HS-overlap at 0.647, is not distinguishable from any other. All nine geometries fall within 0.0141 of one another, with no comparison significant even before correction. This is a genuine null rather than an underpowered one: at $n = 30$ the two-sided Wilcoxon signed-rank test could have returned p as small as 1.9×10^{-9}.

3.8 Finite-shot estimation

All results so far assume noiseless, infinite-shot simulation. Since $\mathrm{tr}(\rho\sigma)$ is the SWAP-test observable, S shots give an unbiased estimate with variance $(1 - k^2)/S$, which we simulate by sampling each unordered Gram entry binomially and projecting back to the positive semi-definite cone (table 11 and figure 10(b)).

The variance gain reported above does *not* translate into a proportionate saving in shots; the reference frame requires more shots to approach its own ceiling, because that ceiling is higher and the distinctions to be resolved are correspondingly finer. Normalising each frame by its own ceiling is in fact misleading, since the sensor frame converges quickly only because it converges to near-chance. The meaningful comparison is absolute: from 10^4 shots per Gram entry upwards, the reference frame exceeds what the sensor frame attains with unlimited shots. Below 10^3 shots both collapse to chance, so the frame raises the ceiling a well-resourced estimator can reach rather than rescuing a starved one.

Near-ceiling accuracy nonetheless requires 10^5–10^6 shots per Gram entry. A 45-trial subject has of order 10^3 unordered pairs, so a single Gram matrix costs 10^8–10^9 shots, and the IV-2a trial count raises that by a further factor of about forty. The reference frame makes these kernels work; it does not make them cheap.

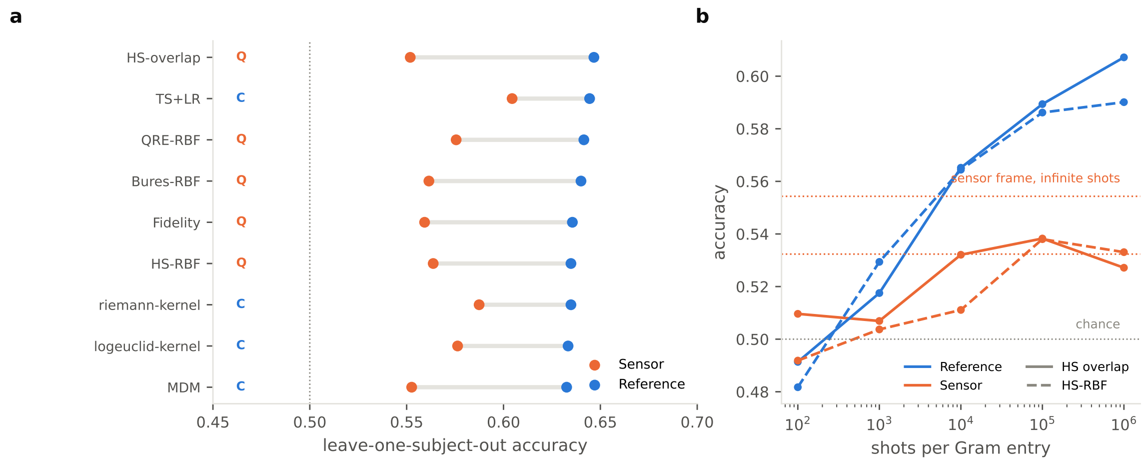

Figure 10. (a) Cross-subject transfer, leave-one-subject-out. Recentring helps every method and helps the quantum kernels (Q) most, but in the reference frame all nine geometries are indistinguishable. (b) Accuracy against shot budget. The reference frame needs more shots to approach its own ceiling because that ceiling is higher, yet from 10^4 shots per Gram entry it already exceeds what the sensor frame attains with unlimited shots.

3.9 Computational cost

The cost asymmetry is substantial. classical/logvar+LDA reached its accuracy in 0.046 s per subject, while quantum/Bures-RBF-SVM required 12.4 s, a factor of 272, for no accuracy gain. This is measured under noiseless, infinite-shot simulation; on hardware, estimating $\mathrm{tr}(\rho\sigma)$ to precision ϵ costs $O(\epsilon^{-2})$ shots per Gram entry, and the concentration results above imply that the required ϵ shrinks exponentially with register size.

4 Discussion

4.1 Interpretation

Under matched tuning, matched features and paired statistics, quantum kernels did not improve motor-imagery decoding. That conclusion is consistent with the broader QML benchmarking literature [16], but the route by which we reach it differs from a bare negative result in a way we think matters, and it passes through a stage at which the opposite conclusion looked correct.

Evaluated in the sensor frame, the density-matrix kernels, the most principled construction here, requiring no dimensionality reduction and mapping onto a hardware-native SWAP test, performed *worst* of all pipelines, and section 3.3 appeared to explain why: EEG covariances are mutually so similar that their quantum overlaps saturate near unity. That explanation is incomplete. Concentration is a symptom of a deeper mismatch, and it is largely removable. Uhlmann fidelity, the Bures distance, the Hilbert–Schmidt overlap and the quantum relative entropy are invariant under *unitary* conjugation; EEG's nuisances act by *congruence*. The affine-invariant Riemannian metric is invariant under that whole group, which is the structural reason tangent-space decoding is the classical state of the art. Comparing the two families in the sensor frame therefore measures, in part, a difference of invariance groups rather than a difference of geometries.

Correcting this ((6)) relieves concentration by 4.7–9.5×, lifts every kernel on both datasets, and reverses the sign of the headline comparison. Taken alone, that is a publishable claim of quantum advantage at $p = 0.059$, replicated on a second dataset, with the effect *growing* on the better-powered one. We think it is worth being explicit that this is what our own experiments supported at that point, because the only thing standing between it and publication was a control that is not currently standard.

That control, an SPD kernel in the same classifier, same data, same frame, differing only in the metric, removes the claim entirely. Across four settings no quantum kernel separates from its classical twin, and the twin is itself among the strongest pipelines we evaluated. The gain therefore belongs to the reference frame and to the kernel formulation, both of which are available classically and neither of which requires a quantum computer. Once the invariance groups are matched, the quantum geometries are not better; they are not worse either, which is itself informative, since it locates the ceiling precisely where the algebra says it must be.

We draw two lessons. The first is specific: quantum information geometry is a mathematically exact description of EEG covariances that confers no decoding benefit over its classical counterparts, and reports to the contrary should be checked for a frame asymmetry before they are believed. The second is methodological: the dimension-matched control that this literature has begun to adopt is necessary but not sufficient. Its natural analogue for geometric methods, a *metric*-matched control, identical in every respect but the metric, is what distinguishes a claim about quantum structure from a claim about preprocessing.

4.2 Relation to prior work

Our findings are compatible with reports of quantum benefit on EEG [6, 7] only under the interpretation that those benefits arise from the classical components of the hybrid pipelines. Our dimension-matched control speaks directly to this: a plain linear SVM on the same PCA-reduced tangent-space features matched or exceeded every circuit kernel. Where an advantage is reported without such a control, the tangent-space or spatial-filtering front end remains an untested explanation.

4.3 Limitations

Five limitations bound our conclusions. (i) We study two datasets but one binary paradigm, left hand versus right hand; other motor-imagery classes and other BCI paradigms (P300, SSVEP) are untested. The trial-count concern that a single dataset would leave open is addressed by section 3.4. (ii) Our central claim is an equivalence, and equivalence is bounded rather than absolute. Two one-sided tests place the quantum kernels and their classical twins within ±0.032 accuracy of one another across all 20 comparisons (table 9), so we can exclude differences larger than that, but not smaller ones. The TOST implementation is the paired t form and assumes the per-subject differences are approximately normal; at $n = 30$ that is mild, at $n = 9$ it is a real limitation and the IV-2a bounds should be read as indicative. (iii) Our invariance argument concerns congruence. Inter-subject and inter-session shift is dominated by, but not exhausted by, that group; differences in conditioning, effective rank and spectral shape survive recentring, and it remains possible that a quantum divergence weights those residual differences more usefully in some regime we did not probe. (iv) The reference state is a Fréchet mean estimated from training data, so the correction is only as good as that estimate; its markedly larger benefit on IV-2a (+0.164 to +0.187) than on PhysioNet (+0.050 to +0.106) is consistent with this, and implies the correction will be weaker still in genuinely few-trial calibration settings. (v) Simulation is noiseless, and although section 3.8 adds shot noise it models sampling only, not gate error, decoherence or readout bias. All of these favour the quantum side, so none weakens a negative result; they do mean hardware performance would be worse, not better. We also survey a specific family of kernels: deep architectures such as EEGNet [35], and variational or quantum neural-network approaches, are outside our scope.

4.4 What would change the conclusion

Our claim is that the quantum geometries match their classical twins once the invariance groups agree, so what would overturn it is a setting in which a quantum divergence separates from a metric-matched classical comparator. Three seem worth trying, and we state in advance why the first two are the promising ones.

Non-congruence shift is the natural candidate, because it is exactly what our argument does not cover. Cross-session rather than cross-subject transfer is attractive here: the congruence component is milder, so residual differences in conditioning and spectral shape, on which the quantum relative entropy and the Bures distance weight differently from the affine-invariant metric, carry proportionally more of the shift.

Regimes where the reference state is poorly estimated are the second. Our correction depends on a Fréchet mean, and the classical baselines depend on the same estimate; a divergence that degrades more gracefully under a noisy reference would be a real and practically relevant advantage, and few-trial calibration is where it would show.

Finally, and least promisingly on present evidence, larger registers. We observed the sensor-frame kernels deteriorate from three to five qubits while the reference-frame kernels did not, so a systematic sweep to six qubits with both frames would settle whether the concentration of [17] governs the density-matrix family at all once the frame is corrected. We do not expect it to change the metric-matched comparison.

5 Conclusion

Treating EEG covariance matrices as quantum density matrices yields a principled, hardware-mappable family of quantum kernels that requires no dimensionality reduction. Evaluated in the sensor frame, these kernels are the worst pipelines in a controlled benchmark, and appear to be defeated by kernel concentration. That diagnosis is incomplete: the quantum quantities are invariant under unitary conjugation while EEG's nuisances act by congruence, so the comparison was confounded by an invariance mismatch that the classical baselines do not share. Referring each state to a training-set Fréchet mean restores exact affine invariance, relieves concentration by 4.7–9.5$\times$, improves every kernel on both datasets, and reverses the sign of the classical-versus-quantum comparison.

It would be straightforward to stop there and report a quantum advantage. The control that prevents this, a classical SPD kernel identical in every respect but the metric, shows that no quantum kernel is distinguishable from its classical twin in any of four settings, including against filter-bank CSP and in cross-subject transfer, where all nine geometries fall within 0.0141. The improvement is real and useful, and it is entirely classical in origin.

We therefore find no evidence of quantum advantage for motor-imagery decoding, and we locate the ceiling: matched invariance permits parity and nothing more. For practitioners, the reference-state correction is worth adopting on its own terms. For the field, we suggest that a metric-matched control should accompany the dimension-matched one, since without it these same experiments support the opposite conclusion. All code and results are released to support replication.

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Author contributions

Md. Imtiaj Alam Sajin: conceptualisation, methodology, software, formal analysis, investigation, data curation, visualisation, writing – original draft. **Samiul Islam Saif**: software, investigation. **Muhammad Hasibur Rashid Chayon**: supervision, writing – review and editing.

Data availability
The PhysioNet EEG Motor Movement/Imagery Database is publicly available at https://physionet.org/content/eegmmidb/1.0.0/, and BCI Competition IV-2a via MOABB at https://www.bbci.de/competition/iv/. All analysis code, configuration and result files supporting this study, including the per-fold scores behind every table and figure, are openly available at https://github.com/Imtiaj-Sajin/quantaEEG.

Conflicts of interest
The authors declare no conflicts of interest.

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